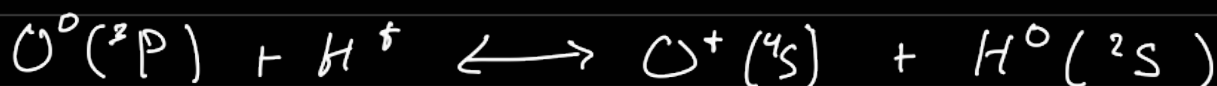


(d) Given that the ionization energy of O is similar to the energies required for H to be ionized. So we expect there to be ionization from the high energy photons from the star.

Along with this is ionization/charge-exchange occurs:

• Collision between O and H:



• Which adds to the abundance of O^+ (OII) and similarly for OIII.

(e) Strong emission lines from OII and OIII occur from forbidden QM transitions.

• OII & OIII interact with H via collisions and excites electrons to a higher state in O. These then fall to a different low energy state:

(OII):

• From this interaction

$$\lambda = 3726 \text{ \AA}$$

$$\lambda = 3729 \text{ \AA}$$

emission lines

(OIII): Much brighter emission than OII

$$\bullet \lambda = 4363 \text{ \AA}$$

$$\lambda = 4959 \text{ \AA}$$

$$\lambda = 5007 \text{ \AA}$$

emission lines

* Because the two OII lines are so close together this gives a very narrow dispersion and this makes it harder to make accurate conclusions of temp

$$\rightarrow \frac{I_{3726}}{I_{3729}} \sim e^{\frac{\Delta E}{kT}}$$

So we use OIII instead.

